

21/09/2026

Recaping to date, I have:

- Set up labbook
- Set up code repository locally
- Received code and basic introduction from Dr Turci
- Partly read introductory papers/books on the topic

Looking forwards, there are 2 more immediate things i need to do:

- continue reading to build up understanding:
 - Second quantisation (Altland and Simons 2010)
 - Introductory paper (Rivas and Huelga 2012)
- Build up main “computational engine”:
 - Look through and understand code Dr Turci has sent me
 - Refactor said code into a convenient utility I can use throughout this project

After i’ve done these i’ll be equipped with the main theoretical and computational tools for this project. This will inform the rest of the investigation of this project.

Having a practical understanding of what this project entails should help me understand the theory “in context”. As such, i think for now its worth examining the code Dr Turci has sent me, in order to start planning the computational engine.

Examining the provided julia code (Y4DissertationCodeMaterials/providedCode/basic-1site-bath-julia.ipynb), it starts by reiterating the maths. Note; explanation recommends finding a good explanation for “wick’s theorem”. Examining my old personal notes for QFT, i recall it being relevant to evaluating the non commuting terms in the Dyson series. That understanding was based on (Schwichtenberg, n.d.) which may be of use. However i wonder if the second quantisation (Altland and Simons 2010) chapter will introduce it in a more contextually relevant manner. Skimming the chapter it seems so... Changed my mind, i’m going to try and quickly skim the chapter with the remainder of this afternoon. I expect the concepts will be familiar, so shouldn’t take too long. Code examination can come tmr.

Reading notes: Second quantisation impressions (Altland and Simons 2010):

Second quantisation seems to be general framework of ladder operators and associated commutation relations; which is a familiar concept. Apparently generalises nicely to N-body quantum physics.

Chapter motivates second quantisation by highlighting computational issues w/many body QM due to indistinguishability, and the exchange of particle numbers.

It then covers the “apparatus of second quantization”.

Compares representation of labelling particles in state individually to simply counting the number of particles in each state (I.E $|1, 1, 1, 1, 2, 2, 3\rangle$ becomes $|4, 2, 1\rangle$). This is the “occupation number representation”. Also note the “Fock space” $\mathbb{F}$

Altland, Alexander, and Ben Simons. 2010. *Condensed Matter Field Theory 2e*. Cambridge University Press. <https://doi.org/https://doi.org/10.1017/CBO9780511789984>.

Rivas, Ángel, and Susana F Huelga. 2012. *Open Quantum Systems. An Introduction*. <https://doi.org/https://doi.org/10.48550/arXiv.1104.5242>.

Schwichtenberg. n.d. *No-Nonsense Quantum Field Theory*.